

VAPO: Value-Model- Based RL for Advanced

Achieving 60.4 on AIME 2024 — 10+ Points
Above DAPO

ByteDance Seed | Qwen2.5-32B Base Model | No SFT Data

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

✗ Value-Model-Free (GRPO/DAPO)

- Trajectory-level advantage assignment
- Uniform credit across all tokens
- Final reward propagated equally
- Subtle step errors → catastrophic failures

VS

✓ Value-Model-Based (VAPO)

- Token-level precise credit assignment
- Lower variance per token vs Monte Carlo
- Value model generalizes across samples
- Critical for multi-step reasoning

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1

Value Model Bias

- Long trajectories compound error
- Bootstrapping amplifies bias
- Initial value estimates unreliable

2

Heterogeneous Sequence Lengths

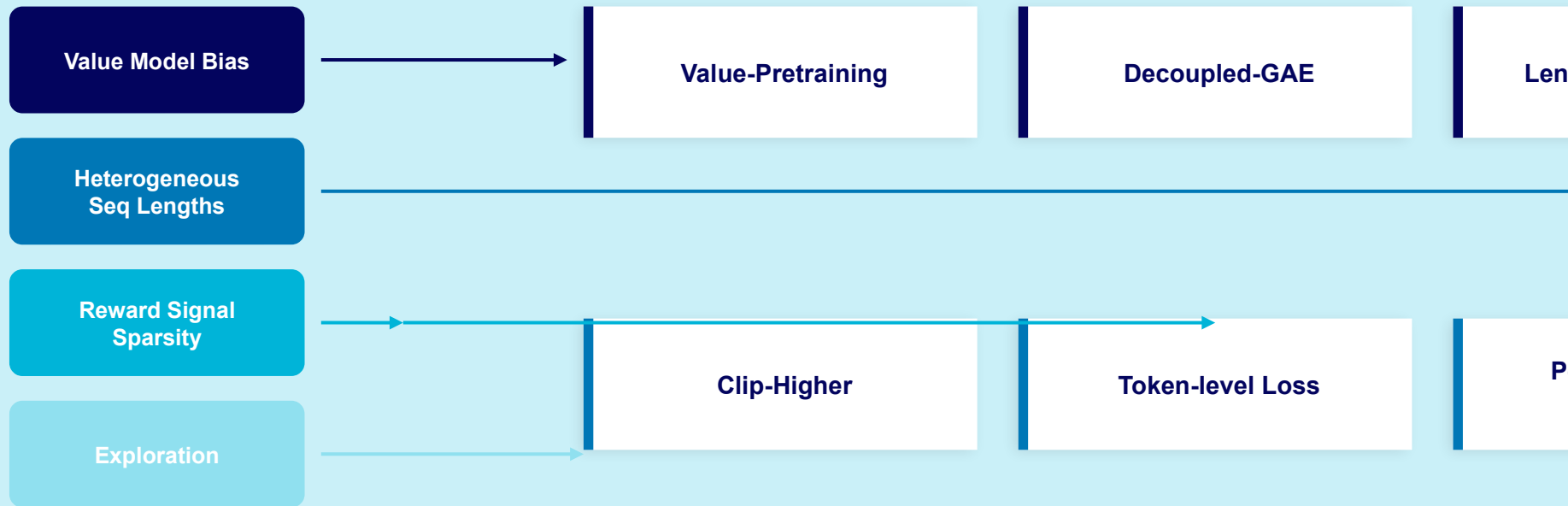
- Short vs long responses differ wildly
- Bias-variance tradeoff varies by length
- One λ doesn't fit all

3

Reward Signal Sparsity

- Only final <eos> token receives reward
- Long reasoning chains = sparse signal
- Early tokens get no learning signal

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



Integrated Framework

Challenge Categories:

Value Bias

Length

Reward

Exploration

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Novel technique for balancing bias-variance tradeoff across response lengths

λ in GAE Controls Bias-Variance Tradeoff

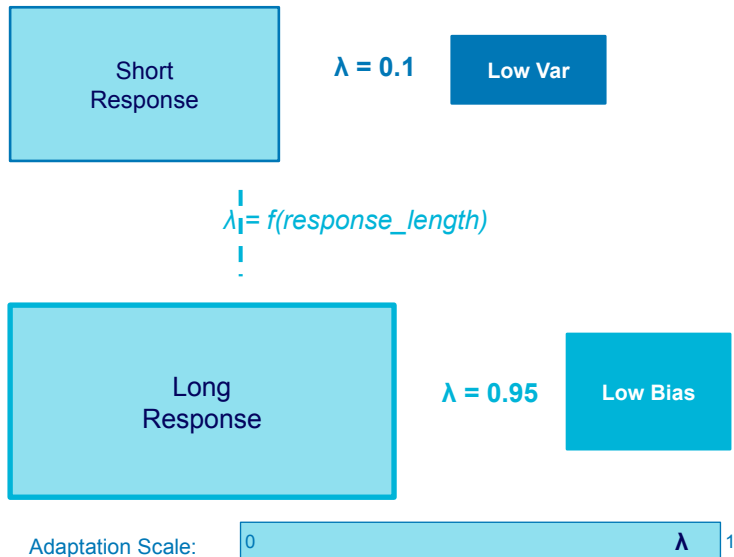
- $\lambda \rightarrow 0$: Lower variance, higher bias
Favors value estimate from bootstrapping
- $\lambda \rightarrow 1$: Higher variance, lower bias
Favors Monte Carlo return estimates

Novel Adaptation: λ Varies with Response Length

- Short responses $\rightarrow$ Lower λ
Lower variance is safe for short horizons
- Long responses $\rightarrow$ Higher λ
Need lower bias for long horizon accumulation

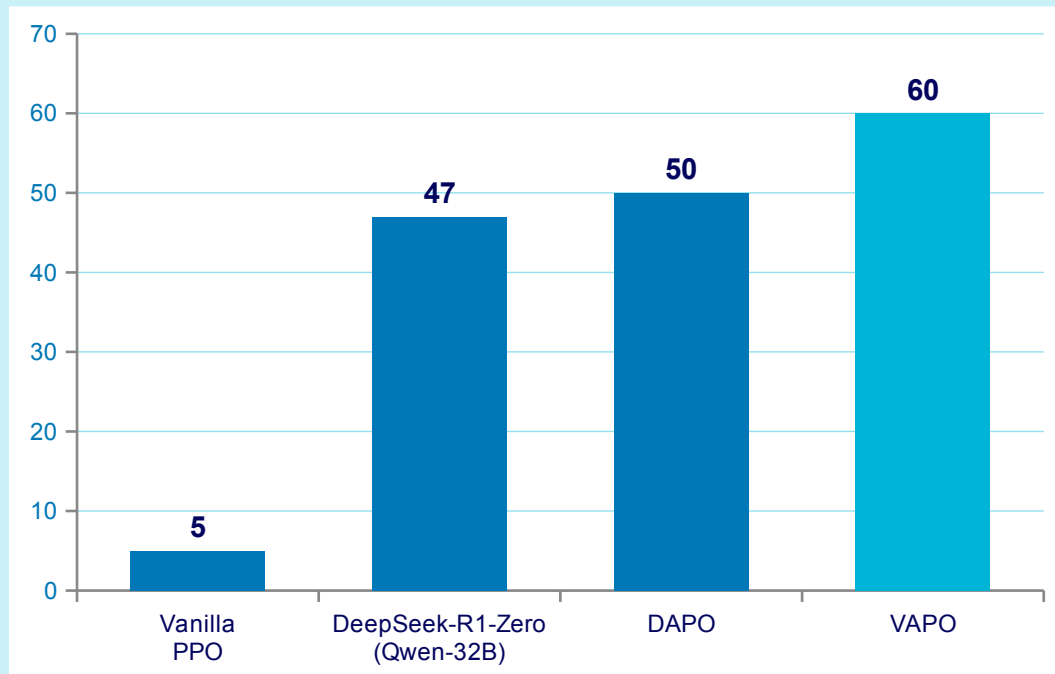
$\lambda = f(\text{length, horizon})$ — adaptive weighting

Length-Adaptive λ Visualization



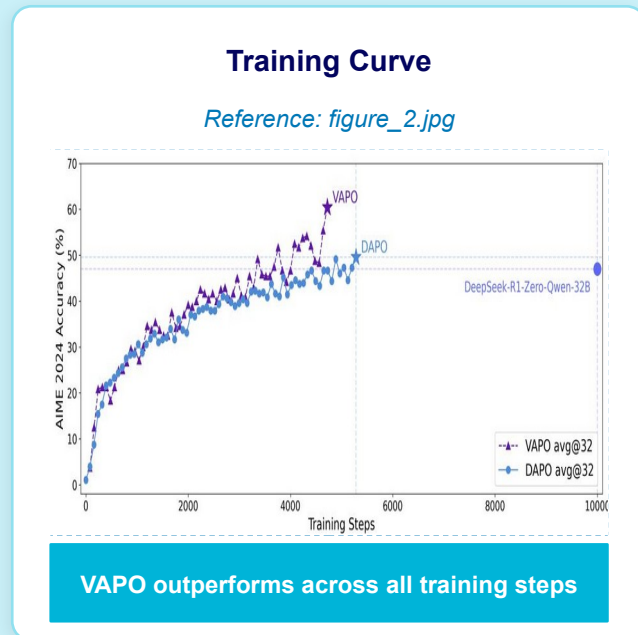
VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

Benchmarking results on competition mathematics

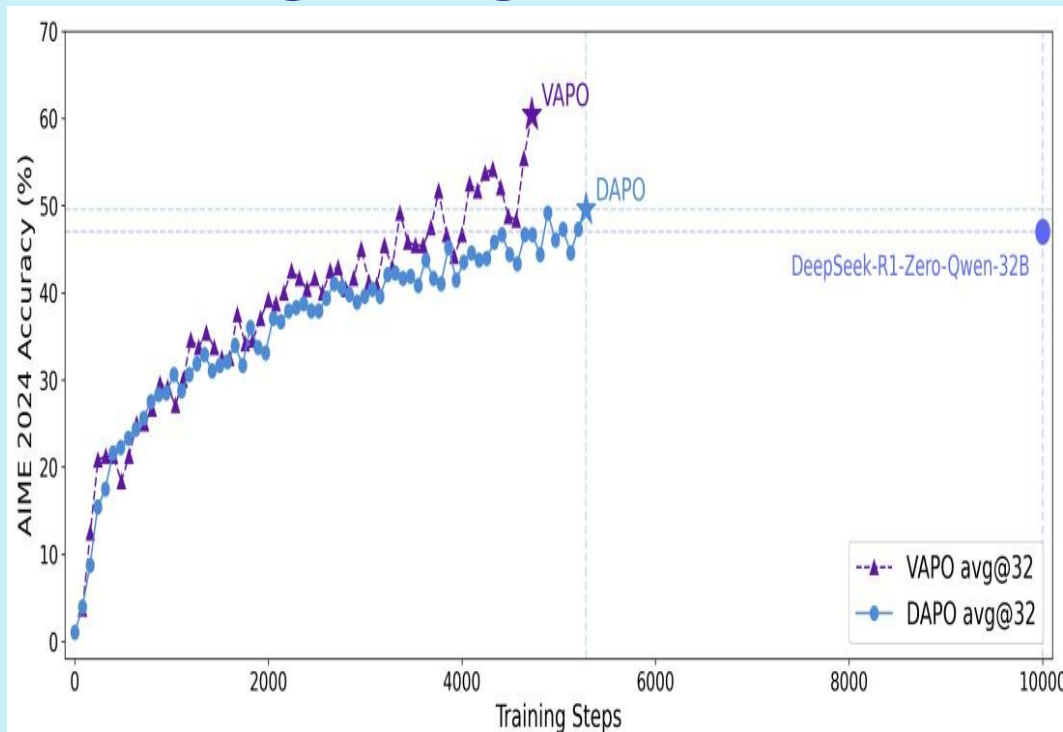


AIME 2024 avg@32

+55 pts vs Vanilla PPO | +13 pts vs DAPO | +10 pts vs DeepSeek-R1-Zero



VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



Key Milestones

VAPO reaches 60%

Step ~4,500

DAPO plateaus at 50%

Step ~5,500

DeepSeek-R1-Zero

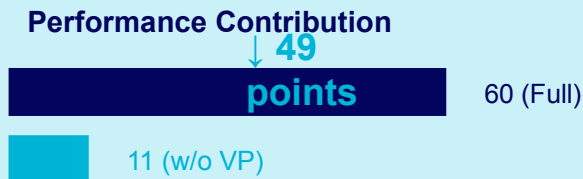
47% @ 10K steps

VAPO consistently outperforms DAPO throughout entire training trajectory

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	Score	Drop
VAPO (full)	60	—
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-adaptive GAE	45	-15
w/o Clip-Higher	46	-14
w/o Token-level Loss	53	-7
w/o Positive Example LM Loss	54	-6
w/o Group-Sampling	55	-5

★ Value-Pretraining is the single most impactful component



Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

- 1 Value-model-based methods can surpass value-free when challenges are addressed
- 2 Length-adaptive GAE is critical for heterogeneous lengths (contributes 15 points)
- 3 Value pretraining is the single most important component (49 point contribution)
- 4 VAPO is stable with zero training crashes across multiple runs
- 5 Implications for future reasoning model development

Value-model-based RL, with proper engineering, delivers superior reasoning performance